Bi-Weekly Status Report

Project Name: Code Sentinel – A Multi-Agent Code Review Platform
Report Period: 17 Sep 2026 – 30 Sep 2026
Prepared By: Sumit Goyal (Orchestration – on behalf of the team)
Date Submitted: 30/09/2026

Project Status Summary

Sprint/Milestone	Orchestration and Agent Services (Security, Style) work packages complete for the MVP; M5 (core pipeline) criteria met in a live run, formal sign-off 09 Oct; M6 mid-semester demo week of 12 Oct
% Complete (overall)	55%

Indicator	Current	Previous	Trend	Status
Overall Project	55%	38%	↑	**On Track**
Schedule	97%	95%	↑	**On Track**
Scope	100%	100%	→	**On Track**
Quality	395 tests (CI green)	223 tests (CI green)	↑	**On Track**
Team Contribution	88%	100%	↓	**On Track**
Dependencies	100%	100%	→	**On Track**

Summary Notes:
This period closed the gap flagged in the last report: the Security and Style agents, which were the critical path to M5, are built, tested and merged, together with the LLM layer and quota-aware provider routing that the plan had scheduled for 05–08 Oct. On 27 Sep a real pull request on the team's playground repository went through the whole pipeline – signed webhook, gateway, orchestrator, both agents in parallel, aggregation – and came back to GitHub as a Code-Sentinel Check Run with annotations and a review with one-click suggested changes. All four MVP criteria in the project plan are therefore met ahead of the 09 Oct milestone, and the GitHub App posting work (planned 19–30 Oct) has been pulled about three weeks forward. The test suite grew from 223 to 395 passing test cases across 50 files. Two items did not go to plan: PostgreSQL migrations slipped a second time (now 07 Oct), and the dashboard shell has not started; neither affects M5, so team contribution stays on track. The LLM provider decision was also revised again, to Groq as primary and Gemini as secondary.

Key Accomplishments in Two Weeks

- **MVP pipeline proven on a real pull request** – webhook from a real GitHub repository, parallel fan-out to both agents with timeouts, de-duplicated report, and a Check Run plus inline suggested changes posted back to the PR (PR #15, PR #17; run log in `plans/github-posting.md`)

- **LLM abstraction layer** (`packages/llm`) – Groq and Gemini providers on plain `fetch` with strict JSON-schema output, Gemini embedder, typed errors and a sliding-window quota budget per provider (PR #9)
- **Quota-aware LLM routing in the orchestrator** – each batch is planned and reserved before fan-out; Groq is used when its quota fits, otherwise Gemini; two bursts of 5 reviews ran with no HTTP 429 (PR #10, PR #14)
- **Shared agent kit** (`packages/agent-kit`) – one HTTP app, config loader, patch parser and review helpers used by every agent, so the three remaining agents start from a working base (PR #11)
- **Security Agent** – 16 CWE-tagged rules and secret detection (patterns plus an entropy check, secrets masked before any LLM call), an LLM pass for issues rules cannot see, and AI-suggested fixes limited to lines added by the PR (PR #12, PR #16)
- **Style Agent** – language dispatcher (JS/TS → ESLint/Prettier, Python → Ruff/Black) running inside a Docker sandbox with no network, and deterministic auto-fixes without an LLM call (PR #13)
- **GitHub App integration** – Octokit client authenticated as the App, Checks API run with summary and annotations, and a PR review with inline comments and one-click suggestions; redelivered webhooks are ignored (PR #15, PR #17)
- **System Design Document updated** – PR review sequence diagram and provider section revised and re-exported to match the implementation

Planned vs. Actual

Task/ Deliverable	Planned Completion	Current Forecast	Actual Status	Notes
Security Agent: SAST rule integration	Sep 14–Sep 18	—	Done (Sep 22)	Four days late (slip reported last period); rules in TypeScript instead of Semgrep/Bandit
Security Agent: secret detection (regex + entropy)	Sep 19–Sep 25	—	Done (Sep 22)	Delivered with the rules; secrets masked before the LLM pass
Style Agent: language dispatcher	Sep 14–Sep 17	—	Done (Sep 23)	Routes by file extension to ESLint or Ruff
Style Agent: Docker sandbox	Sep 18–Sep 21	—	Done (Sep 23)	Linters run in a container with no host or network access
Style Agent: ESLint/Prettier/ Ruff/Black + auto-fix	Sep 22–Sep 25	—	Done (Sep 25)	Fixes limited to added lines; JS hunk parsing fixed after live run
LangGraph state machine design	Sep 28–Sep 30	—	Done	Completed early in the previous period

Task/ Deliverable	Planned Completion	Current Forecast	Actual Status	Notes
Aggregation and de-duplication	Sep 28–Oct 1	—	Done	Extended for LLM line drift (same CWE within 3 lines)
LLM abstraction layer	Oct 5–Oct 6	—	Done (Sep 19)	Pulled forward; used by the Security Agent now
Provider fallback chain	Oct 7–Oct 8	—	Done (Sep 20)	Reduced to Groq → Gemini, routed by quota
PostgreSQL migrations + seed scripts	Sep 15–Sep 16 (rescheduled Sep 30)	Oct 7	In Progress	Slipped a second time; in-memory stores still in use
GitHub App: webhook + Octokit client	Oct 19–Oct 23	—	Done (Sep 25)	Pulled forward about four weeks
GitHub App: Checks API + suggested changes	Oct 24–Oct 30	—	Done (Sep 26)	Pulled forward; verified live on a test PR
End-to-end test on a real PR	Oct 31 / Nov 14	Nov 14	Partly Done	Manual live run passed Sep 27; automated E2E test stays in the testing package
Vector DB (ChromaDB) + embeddings	Oct 19–Oct 24	Oct 24	In Progress	Interface and Gemini embedder ready; Chroma repository still a stub (not MVP scope)
Next.js dashboard shell	Not in plan window	Oct 14	Not Started	Early start promised last period did not happen; planned build is 26 Oct–13 Nov
M5 core pipeline	Oct 9	Oct 9	Criteria met	All four MVP criteria met in the Sep 24 and Sep 27 live runs; sign-off on Oct 9

Project Risks, Issues, & Dependency

Type (ID)	Description	Impact	Owner	Mitigation/Next Step
Issue (I-01)	Security and Style agents not started (from last report)	High	Tej Prakash Tak, Nevil Nandasana	**Closed**: both agents merged and verified on a real PR

Type (ID)	Description	Impact	Owner	Mitigation/Next Step
Issue (I-02)	PostgreSQL migrations and seed scripts slipped again past 30 Sep	Medium	Ayush Soni	Firm date 07 Oct, before the demo; in-memory stores keep the demo working, but data is lost on restart
Risk (R-01)	LLM free-tier quotas: Groq allows about one small batch a minute (8K TPM)	Medium	Sumit Goyal	Quota reserved before fan-out with 80% headroom; overflow goes to Gemini; two bursts of 5 reviews passed with no 429
Risk (R-06)	Demo runs on a local setup (smee relay, Docker on Windows, dev GitHub App); a cold Docker start once returned 503	Medium	Vatsal Kamani	Sandbox health timeout raised to 10 s; write a demo run-book and rehearse on the playground repo before 12 Oct
Risk (R-07)	LLM findings can be wrong or land on the wrong line	Low	Tej Prakash Tak	Findings kept only on added lines; drifted duplicates dropped; low-confidence findings held below the threshold
Risk (R-08)	Dashboard not started; it compresses into late Oct–Nov	Medium	Moksh Mehta, Nevil Nandasana	Start the dashboard shell by 14 Oct against the existing contracts types
Risk (R-05)	Provider decision changed again (Groq primary); older documents describe Gemini only	Low	Sumit Goyal	System Design Document already updated; this report records the change
Dependency (D-01)	GitHub App credentials and webhook secret	Low	Vatsal Kamani	**Closed**: dev App registered with Pull requests and Checks write access

Project Decisions

Decision	Rationale	Impact	Decided By	Date
Groq primary, Gemini secondary (embeddings stay on Gemini)	Groq is faster and has a separate free quota; Gemini covers overflow and failures	Replaces "Gemini only"; routing by quota added to the orchestrator	Team	Sep. 18
Shared `agent-kit` package for all agents	Five agents would otherwise repeat the same HTTP, config and patch code	Logic, Performance and Documentation agents start from a tested base	Parin, Sumit	Sep. 21
Security rules written in TypeScript, not Semgrep/Bandit	No extra toolchain per request; rules only need the added lines	Faster scans; Semgrep can still be added behind the same interface	Tej	Sep. 22
LLM finding within 3 lines of a rule finding with the same CWE is a duplicate	Model line numbers drift (seen in the M5 live check)	No double findings in the report	Tej, Sumit	Sep. 24
Gemini calls may use the time left before the deadline; Groq keeps a 15 s cap	A fallback call must still have time to finish	No reviews cut to rules-only in the second burst	Nevil	Sep. 24
AI fixes posted only when every replaced line was added by the PR	Never suggest edits to code the author did not touch	Safe one-click suggestions on GitHub	Tej	Sep. 26
Post the review before completing the Check Run	The summary can then report any comment GitHub refused	Check Run summary is always accurate	Vatsal	Sep. 26

Project Scope Changes

Change	Action	Reason	Schedule Impact	Approved By
Groq → Gemini fallback (OpenRouter and Ollama stay out)	Added	Groq's free tier is fast but small; Gemini absorbs overflow	None; delivered ahead of the 07–08 Oct task	Team

Change	Action	Reason	Schedule Impact	Approved By
GitHub App posting (Checks API, suggested changes)	Moved earlier	Needed to prove MVP criterion (d) before the demo	−3 weeks (ahead of plan)	Team
AI-suggested fixes in the Security Agent	Added	Makes findings actionable as one-click changes on the PR	Absorbed within the period	Team
Shared agent kit	Added	Removes repeated code across five agents	Reduces effort for the Oct–Nov agent tasks	Team

Upcoming Work (Next Two Week's Plan)

- M5 sign-off on 09 Oct: re-run the full live check on the playground repository and record the result (Sumit Goyal, Vatsal Kamani)
- Prepare the M6 mid-semester demo (week of 12 Oct): demo script, run-book for the local setup, and a rehearsal (whole team)
- PostgreSQL migrations and seed scripts from `schema.sql`, replacing the in-memory stores, by 07 Oct (Ayush Soni)
- Orchestrator failure-path tests (agent timeout, LLM fallback, quota exhaustion) running in CI (Sumit Goyal)
- ChromaDB repository and similarity search behind the existing interface, ahead of the 19–24 Oct window (Durgesh Khushlani)
- Logic Agent prompt design and Performance Agent fixtures on top of the agent kit (Tej Prakash Tak, Parin Makwana)
- Next.js dashboard shell and review list against the gateway contract, started by 14 Oct (Moksh Mehta)
- Start the Test Plan outline for M7 (Ayush Soni, coordinating)

Team Member Contributions

Name: Sumit Goyal
Role: Orchestration – LangGraph workflow & LLM abstraction layer

Tasks Completed	Quota-aware LLM routing and review planning in the orchestrator (PR #10); Groq-primary design documented; M5 live check and GitHub live runs executed and recorded; System Design Document updated for PR posting
Tasks In Progress	M5 sign-off run and orchestrator failure-path tests (by 09 Oct)
Contribution Type	Development / Architecture / Testing / Documentation
Blockers	None
Hours/Capacity	103/200 hrs logged
Evidence	PR #10, PR #14; `services/orchestrator/src/budget`, `plans/llm-agents.md`, `plans/github-posting.md`

Name: Durgesh Khushlani
Role: Orchestration – Aggregation, severity ranking & Vector DB integration

Tasks Completed	Orchestrator HTTP server and review-job controller with job store, so the gateway can drive a real review (PR #8); reviewed and merged the live-check fixes (PR #14)
Tasks In Progress	ChromaDB vector repository and similarity search – interface and embedder ready, about 30% complete
Contribution Type	Development
Blockers	None
Hours/Capacity	101/200 hrs logged
Evidence	PR #8, PR #14; `services/orchestrator/src/jobs`, `src/context`

Name: Tej Prakash Tak
Role: Agent Services – Security Agent & Logic Agent

Tasks Completed	Groq and Gemini LLM client (PR #9); Security Agent with 16 rules, secret detection and LLM pass (PR #12); drifted-finding de-duplication; AI-suggested fixes (PR #16); secret finding wording fix
Tasks In Progress	Logic Agent prompt design (planned start 19 Oct, starting early)
Contribution Type	Development / Testing
Blockers	None
Hours/Capacity	92/200 hrs logged
Evidence	PR #9, PR #12, PR #16, PR #17; `services/security-agent` (43 test cases)

Name: Nevil Nandasana
Role: Agent Services – Style Agent

Tasks Completed	Style Agent: language dispatcher, Docker sandbox and ESLint/Prettier/Ruff/Black with auto-fix (PR #13); Gemini deadline fix (PR #14); JS hunk brace balancing after the live run (PR #15)
Tasks In Progress	Style Agent Docker integration test in CI (currently skipped without Docker)
Contribution Type	Development / Testing
Blockers	None
Hours/Capacity	91/200 hrs logged
Evidence	PR #13, PR #14, PR #15; `services/style-agent` (14 test cases)

Name: Parin Makwana
Role: Agent Services – Performance Agent & Documentation Agent

Tasks Completed	Shared agent kit: HTTP app, config, patch parser and review helpers, plus service-to-service auth moved into `service-kit` (PR #11)
Tasks In Progress	Hot-path and missing-docstring fixtures on top of the agent kit, ahead of the 19 Oct build start
Contribution Type	Development / Architecture
Blockers	None
Hours/Capacity	89/200 hrs logged
Evidence	PR #11; `packages/agent-kit` (17 test cases)

Name: Ayush Soni
Role: Backend & Database – API Gateway & PostgreSQL

Tasks Completed	Shared service kit extracted from the gateway (16 Sep), used by the orchestrator and agents; gateway config and store changes reviewed for the GitHub App client (PR #15); migration plan mapped from `schema.sql` to the in-memory store interfaces
Tasks In Progress	PostgreSQL migrations and seed scripts – slipped from 30 Sep to 07 Oct; Test Plan outline and test-case template for M7
Contribution Type	Development / Testing / Documentation
Blockers	None reported; migrations not yet merged
Hours/Capacity	104/200 hrs logged
Evidence	`packages/service-kit`; `docs/design/schema/schema.sql`

Name: Vatsal Kamani
Role: Integrations & DevOps – Monorepo, GitHub App & Deployment

Tasks Completed	Octokit GitHub App client replacing the stub (PR #15); Checks API run, PR review with inline comments and one-click suggestions (PR #17); dev GitHub App registered
Tasks In Progress	Demo run-book for the local setup (smee, Docker, dev App)
Contribution Type	Development / DevOps
Blockers	None
Hours/Capacity	98/200 hrs logged
Evidence	PR #15, PR #17; `services/gateway/src/github` (105 gateway test cases)

Name: Moksh Mehta
Role: Frontend – Next.js dashboard

Tasks Completed	Reviewed and merged the LLM layer (PR #9); mapped the Check Run summary, findings and coverage fields to the report detail wireframe; drafted the mid-semester demo slides and walkthrough script
Tasks In Progress	Next.js dashboard shell – not started, now due to start by 14 Oct; demo slides to be finalised after the 09 Oct M5 run
Contribution Type	Documentation / Review / Presentation
Blockers	None
Hours/Capacity	94/200 hrs logged
Evidence	PR #9; `docs/design/wireframes`